5.methode of final control inspection in preparation area	Defect is not clear enough with visual control	>	>	The method of detection visually is not sufficient.	The operator cannot easily detect the little deformation of pin by eyes	>	risk not tak the PFMEA

1 Result(s)

Nidec Comment

Lesson Learned Card

LLC Type

Date of LLC: **10/22/2024**

Logistics: ☐ Management: ☐

Product: ☐ Detection: ☑

Occurence: ☑ Process: ☐

1. What was the problem?

What happened?

bent hall pins in connector

Who detected it?

operator

Where was it detected?

on production line

How was it detected?

during production

Why is it a problem?

no possibility to insert the plug

When did it happen?

9/5/2024 11:00:00 PM (your local time)

How many parts ?

3.00

2. What is the cause?

the shape of pin of test on last station on electrical test have impact on the deformation of contact plat (HAll IC pin)

3. Description of the applied modification

BEFORE

the surface of contact between the HAll IC pin and the pin of test is Cylindrical
No detection EOL line with gauge of location of pins
no camera with big screen in control of preparation

AFTER

the surface of contact between the HAll IC pin and the pin of test is Rectangular
Reinforcement of detection localization of pin EOL with gauge in EOL final control
Implementation camera in control preparation station

4. What we learned?

ERADICATION

What is the countermeasure that was put in place to avoid recurrence ?

For occurence
***/change the shape of electrical pin of test for production line CM3**
***/Prepare 30 pcs with 100% passed on gauge in preparation area and passed on electrical test to check the reliability of new pin of test**
For non detection
***/Reinforcement of detection localization of pin EOL with gauge in EOL final control**
***/Implementation camera in control preparation station**

CHECKING

What is the method (who, what, how) for the countermeasures?

Who : quality agent and supervisor
how : visuel
what :
1/For occurence
***/ check the pin of test conformity**
2/For non detection
***/ verification the apply of standard in EOL final control**
***/ verification of apply the standard of method of control with gauge EOL**

SIGNALING

What is the countermeasure supposed to d conformity ?

In case of detection of NOK parts with bent l connector in the production line :
+ alert and block all stock for sorting
+ Alert and block the parts in transit

5. What has been modified?

a. Proces is modified according to Before/ After sheet: ☑

b. All documents have been updated and has been validated with the operators: ☐

c. Procedures and work instructions: ☑

d. Control Plan and PFMEA: ☐

e. Maintenance instructions: ☐

f. Records (start-up checklists, reaction sheets, preventive maintenance...): ☑

i. Pr

j. Opera

An evidance of modification:

Electrical_pin_of_test.pptx

presentation_of_change_the_shape_of_el

Evidence_test_localisation_of_30_pcs_ (1).

presentation_of_gauge_of_final_control_of_pin_location.pptx

6. Potential applicability

LLC Applicability Product: **Brush mounting plate assy**

LLC Applicability Line / process: **CM4**

LLC Applicability Plant: **AVO**

Sorting Approval

Approval of sorting / rework

This document must be sent to Nidec within 24 hour(s) after claim notification

Date of page: Update:

CLAIM ISSUER

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CLAIM RECEIVER

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IMMEDIATE ACTIONS REQUIRED AT NIDEC

PROBLEM DESCRIPTION

bent hall pins in connector

DESCRIPTION OF SORTING/REWORK

QUANTITY AND PART NUMBERS CONCERNED

Parts at Nidec plant: **0.00**
Parts at Nidec logistics platfrom: **0.00**
Parts at Nidec's customer (C1): **0.00**
Parts in transport: **0.00**
Part Number & Description: **Brush mounting plate assy (503-773-99-99)**

AVO

Suspected delivery notes and batches#:

SORTING COMPANY CHOSEN TO DO THE SORTING/REWORK

Company Name:
Company Address: